

Flight Company Performance Analysis Dashboard

Project Overview

This project is a **Power BI–based analytical dashboard** designed to evaluate and compare the performance of multiple airline companies. It simulates a real-world airline analytics scenario by combining operational, financial, and customer-related data to deliver **actionable business insights**.

The dashboard focuses on answering key business questions such as:

- Which airlines generate the highest revenue and profit?
 - How do domestic and international flights compare in terms of profitability?
 - What is the distribution of flight statuses (On-Time, Delayed, Cancelled)?
 - Which cities and routes contribute most to passenger traffic?
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Objectives

- Analyze airline revenue, profit, and operational efficiency
 - Compare airline performance across multiple KPIs
 - Visualize trends using interactive Power BI visuals
 - Demonstrate data modeling, DAX, and storytelling skills
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Dataset Description

The dataset represents **simulated airline industry data** and includes:

Key Tables

- **AirlineCompanies** – Airline master data (Airline Name, Annual Revenue, Fleet Size, Global Ranking, Avg Ticket Price, Employees)
- **FlightData** – Flight-level details (Aircraft Model, Destination, Distance, Costs, Status)
- **ON-TIME** – Aggregated on-time flight performance
- **DomesticProfitFirstClass** – Profit data for domestic first-class flights

Key Metrics

- Revenue (Annual & Total)
- Profit per Year
- Passenger Count
- Flight Count
- Distance (KM)
- Ticket Price Range

Data Modeling

- Star-schema-like structure
 - One-to-many relationships between AirlineCompanies and FlightData
 - Proper use of primary and foreign keys
 - Calculated columns and measures using **DAX**
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Dashboard Pages & Insights

1.Flight Company Performance Analysis Dashboard

Insights:

- Passenger distribution by city
- Revenue comparison across airlines
- Market share based on airline share price
- Airline-wise ticket pricing, baggage allowance, meals, and total revenue

2 . Airline Flight Performance Dataset Dashboard

Insights:

- Revenue ranking by airline
- Flight status distribution (On-Time, Delayed, Cancelled)

3.Decomposition Tree Analysis

Purpose:

- Drill-down analysis of **Annual Revenue** by:
 - Airline Name
 - Flight Type (Domestic / International)
 - Profit per Year
 - Distance
 - Flight Status

This enables root-cause analysis for revenue changes.

4. Summary Tables

- Airline-wise total profit
- Airline-wise total flights and revenue

- Quick executive-level snapshot
 - Profit comparison: Domestic vs International flights
 - KPI cards showing total profit, revenue, airlines count, employees
 - Smart Narrative for automated insights
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Tools & Technologies Used

- **Power BI Desktop** – Data visualization & modeling
 - **DAX** – Measures & calculated insights
 - **SQL (Backend)** – Data preparation & structuring (as per project scope)
 - **Excel / CSV** – Data source (simulated)
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Key Skills Demonstrated

- Data modeling & relationship management
 - Business-focused dashboard design
 - Advanced Power BI visuals (Decomposition Tree, Smart Narrative)
 - KPI development & storytelling
 - Analytical thinking using airline industry use-cases
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